Math for CS & AI: Homework 5

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(Adapted from earlier versions of this course taught by Andrew Yao)

Due Date: Apr 6, 2026

Problem Set

- Exercises 3.3–3.11 from Chapter 3 of the lecture notes:
  https://cloud.tsinghua.edu.cn/d/57ddec714cf74c1cab85/

Writing Guidelines

- Same as the first homework.
- We may deduct up to 3 points for poor writing.
- **Special Note:** Your submission should contain only a single PDF file. Please do not submit other files such as `*.zip` or `*.tex`.

Additional Hints (Experimental)

- We provide additional hints for some of the homework problems in the next few pages.
- If you get stuck on a problem, scroll down to find that problem and read the hints.
- Scroll *very slowly* so that you do not accidentally see hints for problems you have not thought about yet.
- Homework is meant to help you practice your problem-solving skills. It is your job to make sure you have thought enough before reading the hints. Ultimately, there will be no additional hints and no AI tools in the exams.
- Again, **peer learning is strongly encouraged**. Reading hints or getting help from AIs is not the same as learning from your classmate who has thought about the problem and can explain it to you. Make some friends who are willing to discuss problems or even have hotpot with you. Learn from their way of thinking, including both successful and unsuccessful attempts.

The rest of this document contains hints for the homework problems.



Think twice before scrolling down!

Hints

Problem 3.3

- Check Proposition 3.11.
- Use linearity of expectation.

Problem 3.4

- Well, you have to do some brute force.
- Expand the sum and use linearity of expectation.
- Random variables are of zero mean.
- If some X_i only appears once in a term, then it is of zero mean.

Problem 3.5

- Very challenging problem if you have not seen it before!
- For (a), try power functions first.
- For (b), shift the probability mass to

$$p'(\omega) = p(\omega) \cdot \frac{Y(\omega)^q}{\|Y\|_q^q}.$$

- For (c), let

$$A = |X|^p, \quad B = |Y|^q.$$

- For (d), let $\alpha = \|X\|_p$ and $\beta = \|Y\|_p$. Now the problem reduces to

$$\mathbb{E}\left[\left|\frac{X}{\alpha} \cdot \alpha + \frac{Y}{\beta} \cdot \beta\right|^p\right] \leq \alpha^p + \beta^p.$$

Problem 3.6

- For (a), use Cauchy-Schwarz inequality.
- For (b), set $a = \theta\mathbb{E}[X]$. If you still do not get the desired inequality, read the proof of Markov's inequality again.
- For (c), read the proof of Chebyshev's inequality again.

Problem 3.7

- Don't try to apply any concentration bounds here. You need anti-concentration bounds.
- Do the previous problem first.
- Use the Paley–Zygmund inequality to show that for a random walk on a real line, with constant probability, the distance from the origin after t steps is at least $\Omega(\sqrt{t})$.
- Prove that for some $T = \mathcal{O}(n^2)$, the probability of visiting all vertices is at least $1/2$ after T steps.
- Split t steps into t/T blocks of T steps each. We only need one successful block.

Problem 3.8

- Please go through the definitions and properties of the expectation, MGF, rate functions, convex functions.

Problem 3.9

- Use Hoeffding's inequality.

Problem 3.10

- Use McDiarmid's inequality.
- If you get a worse bound, you are applying the inequality to wrong random variables!

Problem 3.11

- Start with $n = 1$ first.
- Read the proof of McDiarmid's inequality.
